

RAUNAK MONDAL

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EDUCATION

Carnegie Mellon University, School of Computer Science

Aug 2025 – May 2029

Bachelor of Science in Computer Science (Artificial Intelligence Focus)

Coursework: Data Structures & Algorithms (15-122, C), Introduction to Computer Systems (15-213), Principles of Functional Programming (15-150, SML), Mathematical Foundations for CS (15-151), Concepts of AI (07-180), Matrices & Linear Transformations (21-241), Multivariable Calculus

EXPERIENCE

Software Engineering Intern | Twinly (AI Personal Operator Startup)

May 2026 – Present

- Porting and enhancing the Electron + React Windows desktop application to achieve feature parity with the macOS SwiftUI version, resolving 15+ cross-platform UI discrepancies across navigation, layout, and component rendering.
- Implemented security hardening measures including input sanitization, secure IPC channels, and authentication flow improvements.
- Integrated CAD file preview rendering and multi-account Gmail support, expanding core product functionality for enterprise users.

Founding Engineer | Smartbite AI (Featured at CES 2026)

Jan 2025 – Present

- Architected and built core product features from inception as a founding engineer, contributing to the live product demo at the Consumer Electronics Show (CES) 2026.
- Developed AI agent workflows for personalized meal planning and real-time grocery optimization, integrating LLM-based recommendation pipelines with nutritional constraint solvers.

Lead Developer | ScottyLabs, Carnegie Mellon University

Aug 2025 – Present

- Managed a team of 10 developers building the GSA Student Housing platform, a full-stack web application enabling CMU students to discover, compare, and review off-campus housing listings.
- Defined technical architecture, delegated feature development across frontend and backend teams, and conducted code reviews.

RESEARCH EXPERIENCE

Student Researcher | University of Hawaii, Hawaii Digital Health Lab

Jun – Aug 2023

- Built an LSTM (Long Short-Term Memory) neural network pipeline to detect autism spectrum markers from behavioral video data of children, achieving automated classification of autism-related behaviors.
- First author of resulting paper accepted to AMIA 2024, evaluating which sequence-based neural architectures yield optimal behavior classification performance.

Research Intern | UC San Diego, Dept. of Computer Science

Jun 2022 – Aug 2023

- Created a labeled dataset to analyze ableism in research literature and social media; implemented an LLM-based ableist language detection mechanism using prompt engineering.
- Co-authored “Are Robots Ready to Deliver Autism Inclusion?” published at ACM CHI 2024, a top-tier Human-Computer Interaction venue.

Student Researcher | UCSD Materials Research Science & Engineering Center (MRSEC)

2022 – 2023

- Presented research on Li-ion battery anode synthesis at the UCSD Student Research Conference; applied machine learning to optimize Atomic Packing Factor (APF) predictions across material candidates.
- Developed a computational Magnetic Self-Assembly Simulation model predicting molecular properties, deployed as an interactive web visualization.

PROJECTS

Research Field Simulator (rfsim) | Flask, React, Neo4j, D3.js, Python

- Designed and built a full-stack web application modeling scientific research dynamics across AI/Robotics, Semiconductors, and Synthetic Biology using OpenAlex bibliometric data and graph-theoretic metrics.
- Implemented k-truss decomposition, spectral clustering, Gaussian Mixture Models, and a Wasserstein GAN pipeline for synthetic heterogeneous graph generation; integrated Neo4j backend with D3.js visualizations and a discrete-event simulation engine.

StudyForge | Chrome Extension, JavaScript, Anthropic API

- Built a Chrome Manifest V3 extension with flashcard and practice exam generation powered by the Anthropic API, supporting customizable study workflows for students.

Pittsburgh Student Renters Network (rent-pgh.com) | Full-Stack Web Platform

- Developing a community platform connecting Pittsburgh-area students with verified rental listings, housing reviews, and lease resources to improve off-campus housing transparency.

PUBLICATIONS

- Mondal, R. et al. “Which Sequence-Based Neural Network Architecture Yields Optimal Performance in Classification of Autism-Related Behavior.” *AMIA 2024*. (First Author)
- Mondal, R. et al. “Are Robots Ready to Deliver Autism Inclusion?” *ACM CHI 2024*. (Co-Author)

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript/TypeScript, HTML/CSS, Standard ML (SML)

AI/ML: PyTorch, TensorFlow, NumPy, LLM prompt engineering, LSTM networks, GANs, scikit-learn

Frameworks & Tools: React, Flask, Electron, Node.js, D3.js, Neo4j, Git, Linux, Docker, REST APIs, OpenAlex API

HONORS & AWARDS

National Merit Finalist (2024–25) • Coca-Cola Scholar Semifinalist (2024–25) • Stanford SUMaC (2024) • HMMT 21st in Guts Round • ARML Top 50 Nationally (2023) • 2nd Place, SoCal Science Olympiad at Caltech (2023) • MATHCOUNTS Volunteer Math Coach (2021–24)